

# Computer Vision Helmet

## Post Graduate Program in Artificial Intelligence and Machine Learning

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# Executive Summary

**SafeGuard Corp evaluated four computer vision models** to develop an automated helmet detection system capable of identifying workers who are compliant or non-compliant with workplace safety regulations. The models included convolutional neural networks, transfer learning approaches, and data augmentation techniques. Each model was assessed using validation accuracy, precision, recall, F1-score, generalization capability, and scalability to determine the most effective solution for real-time safety monitoring.

The evaluation showed that all four models delivered strong classification performance and demonstrated the potential to improve workplace safety through automated monitoring. While the transfer learning and augmented models provided improved feature extraction capabilities, some configurations introduced additional complexity and signs of overfitting without significantly improving validation performance. Their higher computational requirements also reduced deployment efficiency across large-scale environments.

**The Simple Convolutional Neural Network** achieved 89% test accuracy, 84% recall, 86% precision, and an F1-score of 85% on unseen test data, demonstrating strong predictive capability and stable generalization across datasets. Most importantly, the model delivered the highest recall among all evaluated approaches, making it the most effective solution for workplace safety applications where minimizing missed helmet violations is the primary business objective. Its simpler architecture also supports faster deployment, lower computational overhead, and greater scalability, establishing the model as the best overall solution for real-time AI-driven helmet

# Business Problem Overview and Solution Approach

## Problem Overview:

Ensuring workplace safety in hazardous environments such as construction sites and industrial facilities continues to be a major challenge, largely due to the inefficiencies and limitations associated with manual monitoring processes. Traditional monitoring methods are highly dependent on human supervision, making them error-prone, inconsistent, and difficult to scale across large operations.

To address these challenges, there is a need for an **automated computer vision-based helmet detection system** capable of continuously monitoring workers, accurately identifying helmet compliance in real time, and improving overall workplace safety enforcement while reducing human error and operational inefficiencies.

## Solution Approach:

To address these challenges, SafeGuard Corp is developing an automated image analysis system powered by computer vision and deep learning. **This system is designed to detect whether workers are wearing safety helmets in real time**, enabling proactive enforcement of workplace safety regulations without relying on human observation.

The proposed solution leverages a binary image classification model designed to automatically distinguish between helmet-compliant and non-compliant workers in hazardous environments. By enabling real-time automated helmet compliance monitoring at scale, the system significantly reduces the human error and inconsistency associated with manual safety oversight.

# Data Overview

## Data Description

### Data set

- Dataset: 4125 images.
  - With Helmet: 3161 images
  - Without Helmet: 964 images

### Characteristics

- **Variations in Conditions:** Images include diverse environments such as construction sites, factories, and industrial settings, with variations in lighting, angles, and worker postures to simulate real-world conditions.
- **Worker Activities:** Workers are depicted in different actions such as standing, using tools, or moving, ensuring robust model learning for various scenarios.

### Images and Labels

- Images: 4125
- Resolution: 200 x 200
- Color Channels: 3 Channels (RGB)
- Labels: 4125



```
print(images.shape)
print(labels.shape)

... (4125, 200, 200, 3)
    (4125, 1)
```

# Exploratory Data Analysis

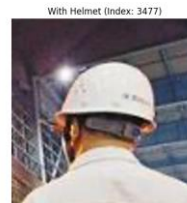
## Random Plot from both Classes

```
num_samples_to_plot = 5

random_helmet_idx = random.sample(helmet_indices, num_samples_to_plot)
random_no_helmet_idx = random.sample(no_helmet_indices, num_samples_to_plot)
```

Label 1: With Helmet

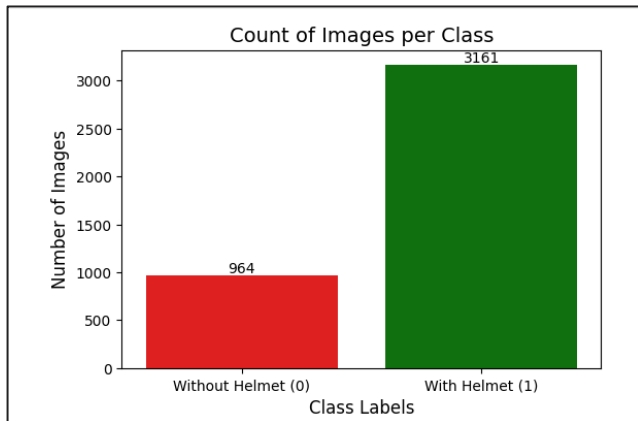
Label 0: Without Helmet



Label 1

Label 0

## Class Imbalance



- The dataset analysis identified a **significant class imbalance** within the helmet detection training data, consisting of 3,161 “With Helmet” images compared to only 964 “Without Helmet” images. This uneven distribution increases the likelihood that the convolutional neural network (CNN) will become biased toward the majority class, potentially favoring “With Helmet” predictions during model training and inference.
- Transfer Learning with VGG-16 combined with FFNN layers and data augmentation** can help address the class imbalance challenge in the helmet detection dataset. VGG-16 leverages pre-trained ImageNet features to improve image recognition, while the FFNN layers adapt these patterns specifically for helmet compliance classification.
- This approach can improve minority-class detection, enhance generalization, and reduce overfitting, resulting in a more reliable workplace safety monitoring solution.

# Data Pre-processing

## Two Step Data Split

Splitting the dataset

```
X_train, X_temp, y_train, y_temp = train_test_split(np.array(images), labels, test_size=0.3, random_state=42, stratify=labels)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, random_state=42, stratify=y_temp) #Complete the
```

```
print(X_train.shape, y_train.shape) #Complete the code to print the shape of the train data
print(X_val.shape, y_val.shape) #Complete the code to print the shape of the validation data
print(X_test.shape, y_test.shape) #Complete the code to print the shape of the test data
```

```
(2887, 200, 200, 3) (2887, 1)
(619, 200, 200, 3) (619, 1)
(619, 200, 200, 3) (619, 1)
```

The data set was split into 30% test and 70% training

The train data set was then split into 15% validation and 15% left for training

Validation data is required to help tune the model without touching the test data thus preventing data leaks.

- ❑ Training Set: (2887 Images)
- ❑ Validation Set: (619 Images)
- ❑ Test Set: (619 Images)

## Data Normalization

```
X_train_normalized = X_train.astype('float32')/255
X_val_normalized = X_val.astype('float32')/255 #C
X_test_normalized = X_test.astype('float32')/255
```

The dataset preprocessing phase included image normalization, where all pixel values were scaled from a range of 0–255 to 0–1 using float32 conversion and division by 255. This standardization improves CNN training efficiency, accelerates convergence, and enhances model stability during learning.

# Model 1: Convolutional Neural Network from Scratch

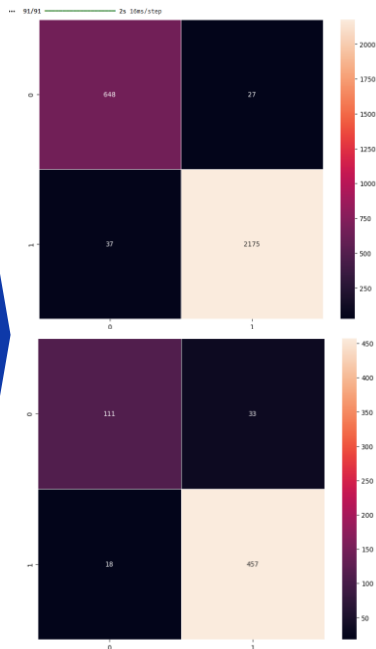
## Summary

Model: "sequential\_1"

| Layer (type)                   | Output Shape         | Param # |
|--------------------------------|----------------------|---------|
| conv2d_3 (Conv2D)              | (None, 200, 200, 16) | 448     |
| max_pooling2d_2 (MaxPooling2D) | (None, 50, 50, 16)   | 0       |
| conv2d_4 (Conv2D)              | (None, 50, 50, 64)   | 9,280   |
| max_pooling2d_3 (MaxPooling2D) | (None, 25, 25, 64)   | 0       |
| conv2d_5 (Conv2D)              | (None, 25, 25, 128)  | 73,856  |
| flatten_1 (Flatten)            | (None, 80000)        | 0       |
| dense_2 (Dense)                | (None, 4)            | 320,004 |
| dense_3 (Dense)                | (None, 1)            | 5       |

Total params: 403,593 (1.54 MB)  
Trainable params: 403,593 (1.54 MB)  
Non-trainable params: 0 (0.00 B)

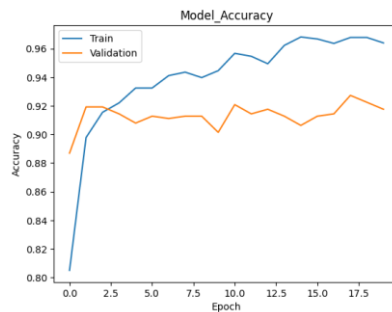
## Confusion Matrix



## Performance Metrics

### Training

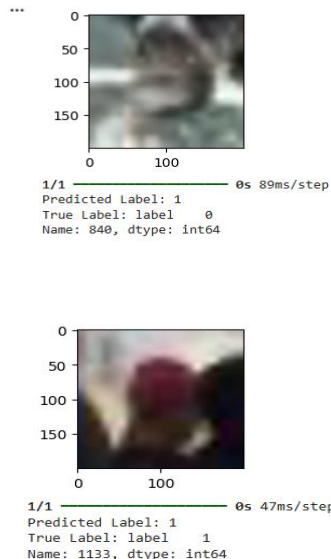
- Accuracy: 0.97
- Precision: 0.96
- Recall: 0.97
- F1: 0.96



### Validation

- Accuracy: 0.91
- Precision: 0.89
- Recall: 0.86
- F1: 0.88

## Visualization



- The custom CNN architecture leverages multiple convolutional and max-pooling layers with progressively increasing filters (16, 64, and 128) to capture increasingly complex image features, followed by flatten and dense layers for final binary classification.
- The model demonstrates strong image classification capability, achieving 91% validation accuracy with balanced precision and recall, indicating reliable real-world performance and stable generalization despite minor overfitting.

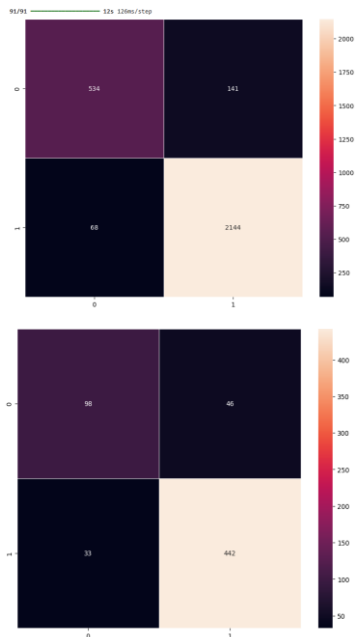


# Model 2: Transfer Learning with VGG-16 (Base)

## Summary

| Layer (type)                            | Output Shape          | Param #   |
|-----------------------------------------|-----------------------|-----------|
| input_layer_2 (InputLayer)              | (None, 200, 200, 3)   | 0         |
| block1_conv1 (Conv2D)                   | (None, 200, 200, 64)  | 1,792     |
| block1_conv2 (Conv2D)                   | (None, 200, 200, 64)  | 36,928    |
| block1_pool (MaxPooling2D)              | (None, 100, 100, 64)  | 0         |
| block2_conv1 (Conv2D)                   | (None, 100, 100, 128) | 73,856    |
| block2_conv2 (Conv2D)                   | (None, 100, 100, 128) | 147,584   |
| block2_pool (MaxPooling2D)              | (None, 50, 50, 128)   | 0         |
| block3_conv1 (Conv2D)                   | (None, 50, 50, 256)   | 295,168   |
| block3_conv2 (Conv2D)                   | (None, 50, 50, 256)   | 590,880   |
| block3_conv3 (Conv2D)                   | (None, 50, 50, 256)   | 590,880   |
| block3_pool (MaxPooling2D)              | (None, 25, 25, 256)   | 0         |
| block4_conv1 (Conv2D)                   | (None, 25, 25, 512)   | 1,180,160 |
| block4_conv2 (Conv2D)                   | (None, 25, 25, 512)   | 2,359,808 |
| block4_conv3 (Conv2D)                   | (None, 25, 25, 512)   | 2,359,808 |
| block4_pool (MaxPooling2D)              | (None, 12, 12, 512)   | 0         |
| block5_conv1 (Conv2D)                   | (None, 12, 12, 512)   | 2,359,808 |
| block5_conv2 (Conv2D)                   | (None, 12, 12, 512)   | 2,359,808 |
| block5_conv3 (Conv2D)                   | (None, 12, 12, 512)   | 2,359,808 |
| block5_pool (MaxPooling2D)              | (None, 6, 6, 512)     | 0         |
| Total params: 14,714,688 (56.13 MB)     |                       |           |
| Trainable params: 14,714,688 (56.13 MB) |                       |           |
| Non-trainable params: 0 (0.00 B)        |                       |           |

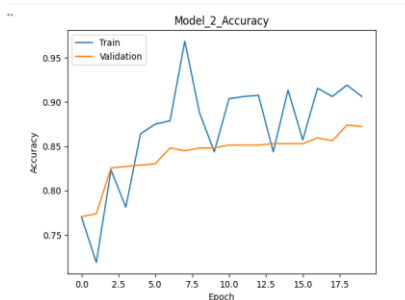
## Confusion Matrix



## Performance Metrics

### Training

- Accuracy: 0.92
- Precision: 0.91
- Recall: 0.88
- F1: 0.89



### Validation

- Accuracy: 0.82
- Precision: 0.82
- Recall: 0.80
- F1: 0.81

## Visualization



- The transfer learning model leverages the VGG-16 architecture, which consists of multiple deep convolutional blocks with progressively increasing filters (64, 128, 256, and 512) and max-pooling layers designed to extract highly detailed hierarchical image features.
- Using pre-trained ImageNet weights, the model achieves 82% validation accuracy with balanced precision, recall, and F1-score, benefiting from strong feature extraction capabilities. However, the gap between training and validation results indicates increased overfitting - additional regularization, fine-tuning, and optimization strategies are needed.

# Model 3: Transfer Learning with VGG-16 (Base + FFNN)

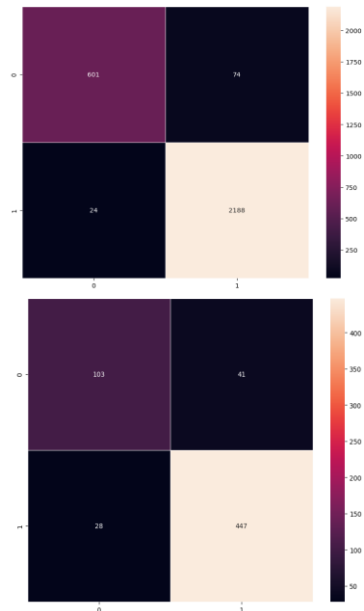
## Summary

Model: "sequential\_4"

| Layer (type)        | Output Shape      | Param #    |
|---------------------|-------------------|------------|
| vgg16 (Functional)  | (None, 6, 6, 512) | 14,714,688 |
| flatten_4 (Flatten) | (None, 18432)     | 0          |
| dense_6 (Dense)     | (None, 256)       | 4,718,848  |
| dropout (Dropout)   | (None, 256)       | 0          |
| dense_7 (Dense)     | (None, 128)       | 32,896     |
| dense_8 (Dense)     | (None, 1)         | 129        |

Total params: 19,466,561 (74.26 MB)  
Trainable params: 4,751,873 (18.13 MB)  
Non-trainable params: 14,714,688 (56.13 MB)

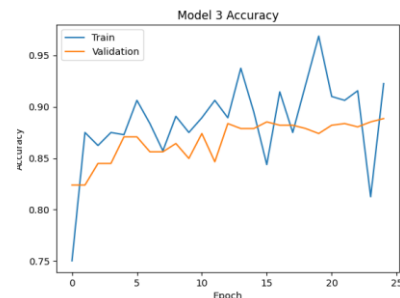
## Confusion Matrix



## Performance Metrics

### Training

- Accuracy: 0.96
- Precision: 0.96
- Recall: 0.93
- F1: 0.95



### Validation

- Accuracy: 0.88
- Precision: 0.85
- Recall: 0.82
- F1: 0.83

## Visualization



Training

Validation

- The transfer learning model combines the pre-trained VGG-16 convolutional base with a custom Feedforward Neural Network (FFNN) classifier, using dense and dropout layers to enhance feature learning and improve classification performance.
- Leveraging ImageNet-trained feature extraction, the model achieves 88% validation accuracy with balanced precision, recall, and F1-score, showing improved performance and generalization over the base VGG-16 model. Although some overfitting remains, the added FFNN layers enhance higher-level feature learning and create a more robust computer vision solution.

# Model 4: Transfer Learning with VGG-16 (Base + FFNN + Data Aug)

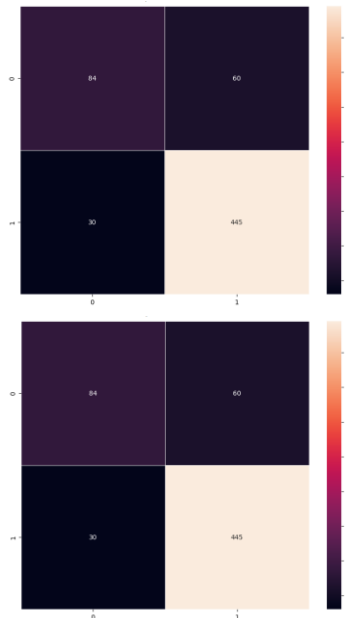
## Summary

Model: "sequential\_6"

| Layer (type)        | Output Shape      | Param #    |
|---------------------|-------------------|------------|
| vgg16 (Functional)  | (None, 6, 6, 512) | 14,714,688 |
| flatten_6 (Flatten) | (None, 18432)     | 0          |
| dense_12 (Dense)    | (None, 256)       | 4,718,848  |
| dropout_2 (Dropout) | (None, 256)       | 0          |
| dense_13 (Dense)    | (None, 128)       | 32,896     |
| dense_14 (Dense)    | (None, 1)         | 129        |

Total params: 19,466,561 (74.26 MB)  
Trainable params: 4,751,873 (18.13 MB)  
Non-trainable params: 14,714,688 (56.13 MB)

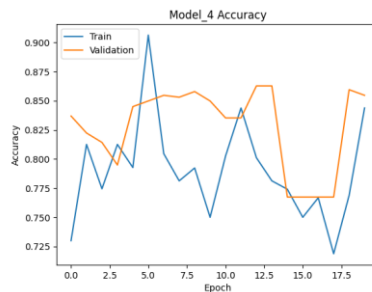
## Confusion Matrix



## Performance Metrics

### Training

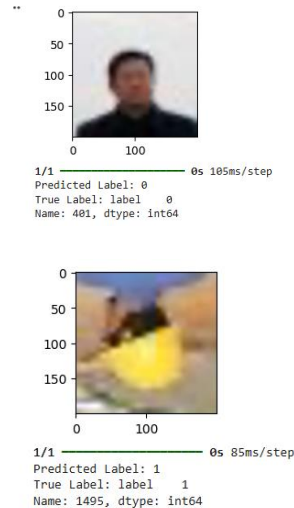
- Accuracy: 0.89
- Precision: 0.86
- Recall: 0.84
- F1: 0.85



### Validation

- Accuracy: 0.85
- Precision: 0.80
- Recall: 0.76
- F1: 0.77

## Visualization



- The transfer learning model combines the pre-trained VGG-16 convolutional base with a custom FFNN classifier and data augmentation techniques to improve model robustness and increase training data variability.
- Using convolutional feature extraction with dense and dropout layers, the model achieves 85% validation accuracy with balanced precision, recall, and F1-score, though performance is lower than the previous VGG-16 + FFNN model. While data augmentation improves generalization and reduces overfitting, the lower validation metrics indicate further tuning is needed to better preserve critical image features and optimize performance.

# Model Performance Comparison and Final Model Selection

models\_train\_comp\_df

|           | Best Performing Model                     |               |                    |                             |  |
|-----------|-------------------------------------------|---------------|--------------------|-----------------------------|--|
|           | Simple Convolutional Neural Network (CNN) | VGG-16 (Base) | VGG-16 (Base+FFNN) | VGG-16 (Base+FFNN+Data Aug) |  |
| Accuracy  | 0.977832                                  | 0.927607      | 0.966055           | 0.897471                    |  |
| Recall    | 0.971637                                  | 0.880185      | 0.939760           | 0.840961                    |  |
| Precision | 0.966862                                  | 0.912668      | 0.964443           | 0.865210                    |  |
| F1 Score  | 0.969221                                  | 0.894930      | 0.951356           | 0.852102                    |  |

## Model Objective

High Recall = Fewer  
False Negatives

Failing to detect a  
worker not wearing a  
helmet creates a  
safety risk

models\_valid\_comp\_df

|           | Simple Convolutional Neural Network (CNN) | VGG-16 (Base) | VGG-16 (Base+FFNN) | VGG-16 (Base+FFNN+Data Aug) |  |
|-----------|-------------------------------------------|---------------|--------------------|-----------------------------|--|
| Accuracy  | 0.917609                                  | 0.872375      | 0.888530           | 0.854604                    |  |
| Recall    | 0.866469                                  | 0.805541      | 0.828165           | 0.760088                    |  |
| Precision | 0.896559                                  | 0.826915      | 0.851122           | 0.809015                    |  |
| F1 Score  | 0.880169                                  | 0.815346      | 0.838720           | 0.779663                    |  |

High Recall = Fewer  
False Negatives

Failing to detect a  
worker not wearing a  
helmet creates a  
safety risk

The Simple Convolutional Neural Network (CNN) was selected as the final model after achieving the highest overall validation performance across accuracy, recall, precision, and F1-score compared to all transfer learning approaches

# Final Model Selection: Simple Convolutional Neural Network

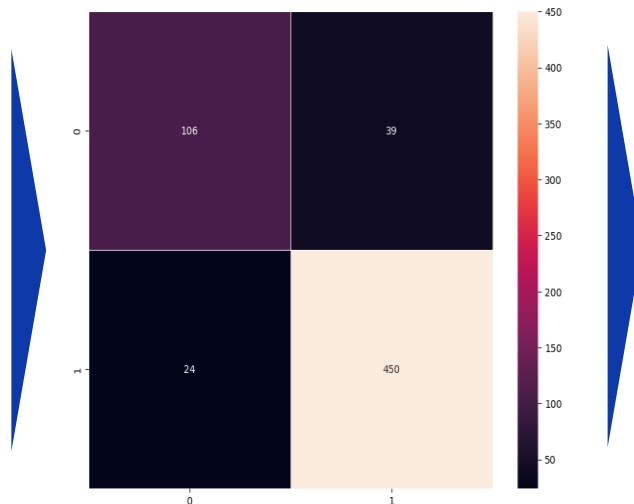
## Summary

Model: "sequential\_1"

| Layer (type)                   | Output Shape         | Param # |
|--------------------------------|----------------------|---------|
| conv2d_3 (Conv2D)              | (None, 200, 200, 16) | 448     |
| max_pooling2d_2 (MaxPooling2D) | (None, 50, 50, 16)   | 0       |
| conv2d_4 (Conv2D)              | (None, 50, 50, 64)   | 9,280   |
| max_pooling2d_3 (MaxPooling2D) | (None, 25, 25, 64)   | 0       |
| conv2d_5 (Conv2D)              | (None, 25, 25, 128)  | 73,856  |
| flatten_1 (Flatten)            | (None, 80000)        | 0       |
| dense_2 (Dense)                | (None, 4)            | 320,004 |
| dense_3 (Dense)                | (None, 1)            | 5       |

Total params: 483,593 (1.54 MB)  
Trainable params: 483,593 (1.54 MB)  
Non-trainable params: 0 (0.00 B)

## Confusion Matrix



## Performance Metrics

- **Accuracy: 0.89**
- **Recall: 0.84**
- **Precision: 0.86**
- **F1: 0.85**

- The Simple Convolutional Neural Network (CNN) emerged as the best overall performing model for the helmet detection solution, delivering the strongest balance of accuracy, recall, precision, and F1-score on unseen test data.
- The model achieved 89% test accuracy, 84% recall, 86% precision, and an F1-score of 85%, demonstrating strong real-world predictive capability and stable generalization across datasets
- Most importantly, the model produced the highest recall among all evaluated approaches, making it the most suitable solution for workplace safety applications where minimizing missed helmet violations is the primary business objective.

# Recommendations

- **Deploy the Simple Convolutional Neural Network (CNN) as the Primary Model**

The Simple CNN demonstrated the strongest overall balance of accuracy, recall, precision, and F1-score while maintaining lower computational complexity and faster deployment capability. Its high recall performance makes it the most suitable solution for workplace safety monitoring where minimizing missed helmet violations is critical.

- **Address Class Imbalance Through Data Augmentation and Class Weighting**

The dataset contains significantly fewer “Without Helmet” images, increasing the risk of model bias toward compliant workers. Expanding minority-class samples through augmentation and applying class weights during training can improve non-compliance detection performance.

- **Implement Real-Time Monitoring and Alerting Capabilities**

Integrating the model into live camera systems can enable automated real-time helmet compliance detection, reducing dependence on manual monitoring and improving workplace safety enforcement responsiveness.

## **Expand Dataset Diversity and Volume**

Future training datasets should include additional lighting conditions, camera angles, worker positions, safety equipment variations, and environmental conditions to improve model robustness and real-world generalization.

- **Introduce Continuous Model Monitoring and Retraining**

Model performance should be continuously monitored after deployment to detect performance drift. Periodic retraining using newly collected workplace images will help maintain long-term accuracy and reliability.



**Happy Learning !**

